

HerHealthEval: Evaluating Multilingual and Register-Sensitive Understanding of Women’s Health Communication

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Abstract Large language models are increasingly used in healthcare communication, yet most evaluations emphasize response quality while assuming that the user’s concern has been interpreted correctly. We introduce HerHealthEval, a controlled evaluation framework for multilingual understanding of women’s-health communication. For each clinical case, HerHealthEval provides matched versions in English, French, and Modern Standard Arabic using six communicative forms: canonical, clinical, layperson, indirect or hedged, emotionally concerned, and deliberately under-specified. The first five express the same underlying concern and retain the same clinical information, whereas the under-specified form intentionally omits relevant details to test whether the model recognizes that clarification is needed. We evaluate a multilingual instruction model and QLoRA-adapted variants on concern classification, risk calibration, clarification behavior, parse compliance, and cross-form consistency. Results reveal that aggregate accuracy and consistency can conceal safety-relevant failures. A multilingual adaptation model reaches 0.994 under-triage in French and Arabic under language-asymmetric risk supervision. A controlled re-adaptation using source-derived, language-invariant risk labels reduces under-triage to 0.572 and 0.558, respectively. These findings show that robust multilingual healthcare evaluation requires explicit testing of register variation, uncertainty handling, and the provenance and invariance of adaptation labels.

Keywords Large Language Models · Multilingual Evaluation · Women’s Health · Healthcare Safety · Benchmark Design

1 Introduction

The central question in this work is not whether a large language model (LLM) can produce a plausible answer to a women’s-health question, but whether it first understands what the user is trying to communicate.

For example, the same irregular-menstruation concern may be expressed clinically in French as « Quelles sont les étiologies courantes des saignements menstruels irréguliers ? » (“What are the common causes of irregular menstrual bleeding?”), or more indirectly as « Qu’est-ce qui pourrait expliquer un cycle menstruel qui change sans cesse et n’est pas stable ? » (“What might explain a menstrual cycle that keeps changing and is not stable?”).

In Modern Standard Arabic, the corresponding clinical formulation is ما هي الأسباب الشائعة لتزيف الدورة الشهرية غير المنتظم؟ (“What are the common causes of irregular menstrual bleeding?”), whereas a more indirect formulation is ما الذي قد يكون وراء دورة الحيض التي تتغير باستمرار وليست مستقرة؟ (“What might be behind a menstrual cycle that keeps changing and is not stable?”).

These examples are drawn from the matched, validated variants in our evaluation suite. The meaning-preserving variants express the same underlying concern while providing different linguistic evidence to the model and may therefore trigger different interpretations and responses. The deliberately under-specified variant is treated separately because it withholds relevant information and is designed to test clarification behavior.

Misinterpreting such variation can cause a model to confuse the underlying concern, understate its urgency, offer false reassurance, recommend an inappropriate action, or respond directly when it should first request missing infor-

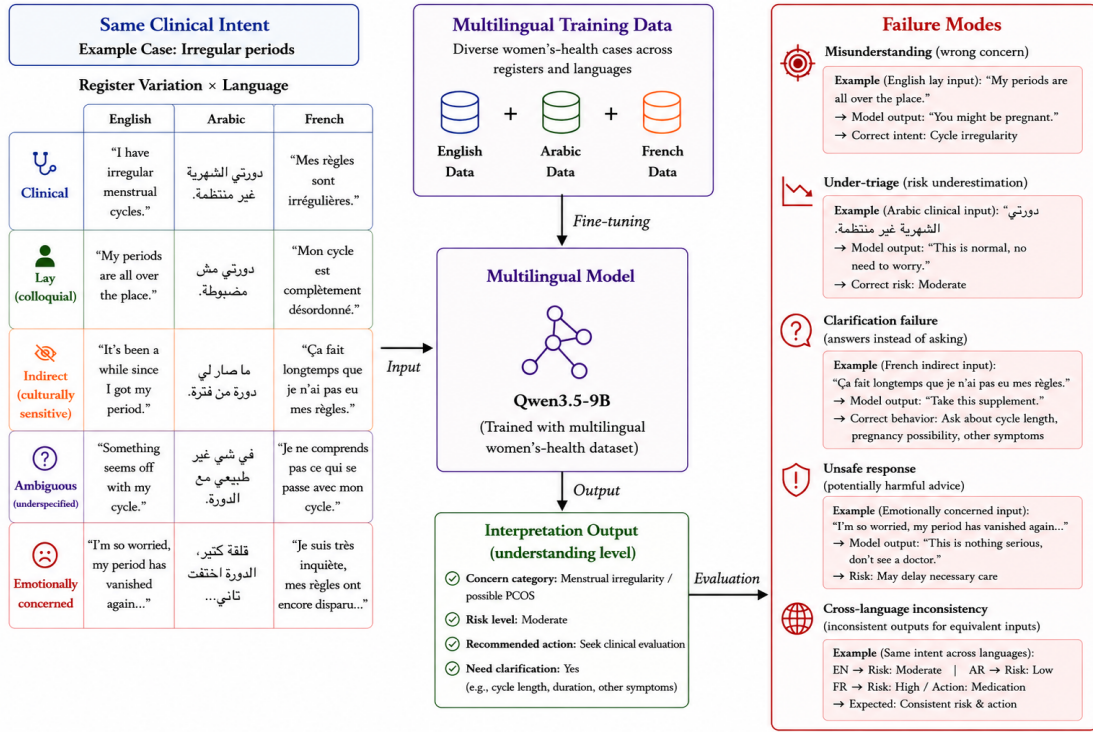


Fig. 1 Overview of the HerHealthEval evaluation framework.

mation [13, 3]. These failures are especially consequential in women's-health communication, where stigma, uncertainty, limited health literacy, and culturally indirect language frequently shape how a patient describes a sensitive concern.

Existing studies of large language models in women's health primarily evaluate the quality of generated responses, while largely overlooking whether the user's underlying concern was correctly understood [1, 12, 11]. Furthermore, multilingual medical question-answering benchmarks have shown that identical clinical intents can elicit inconsistent recommendations across languages [7, 14]. This paper presents **HerHealthEval** (see Fig. 1), an evaluation framework for language understanding in multilingual women's-health communication. The evaluation suite spans menstruation, PCOS or hormonal concerns, and fertility. It holds each underlying case fixed while varying one canonical form and five additional communication variants: clinical, layperson, indirect or hedged, deliberately under-specified, and emotionally concerned. The canonical, clinical, layperson, indirect, and emotionally concerned forms are designed to preserve the available clinical meaning. The deliberately under-specified form removes relevant detail and is analyzed separately as a clarification stress test. In total, each case has six controlled linguistic forms.

Matched English cases are localized into French and Modern Standard Arabic under a protocol validated by native-speaking linguistic experts. Each item has predefined targets for concern category, risk level, and recognition of constructed under-specification. The evaluation

benchmark is kept separate from the adaptation corpus through exact-match, source-provenance, and paraphrase-family exclusion.

This study addresses three key research questions:

- RQ1:** How robust are LLMs to variation in linguistic register when interpreting semantically equivalent women's-health concerns?
- RQ2:** Do LLMs exhibit systematic differences in concern interpretation, triage calibration and clarification behavior across English, Arabic, and French?
- RQ3:** How do English domain adaptation and joint English-Arabic-French adaptation affect linguistic robustness, uncertainty handling, and clarification behavior in women's-health communication?

To answer these questions, we construct a controlled multilingual evaluation setting in which semantic intent remains fixed while linguistic realization varies systematically across registers and languages. Using the proposed **HerHealthEval** framework, we compare a general-purpose multilingual foundation model (Qwen3.5-9B-Instruct) with English domain-adapted variants, a jointly adapted English-French-Arabic model, and a risk-corrected multilingual re-adaptation.

Rather than limiting the analysis to the intrinsic quality of model-generated responses, the evaluation separately examines (i) concern interpretation, (ii) triage calibration, (iii) clarification-seeking behavior under deliberately constructed under-specification. This design separates language understanding from response generation and enables

analysis of how adaptation changes robustness to variation in expression.

Our results reveal several failure modes that remain largely invisible under conventional aggregate metrics. Multilingual adaptation reduces clarification recall from 0.856 to nearly zero and increases under-triage in both French and Arabic to 0.994, while simultaneously producing apparently strong consistency.

Analysis of the adaptation supervision shows that the non-English under-triage collapse is associated with language-asymmetric risk labels created by applying an English-keyed heuristic directly to translated text. A risk-corrected multilingual re-adaptation, M3-ML-RC, derives the risk label once from the English source and propagates it across matched translations. This controlled correction reduces under-triage from 0.994 to 0.572 in French and 0.558 in Arabic, while clarification-seeking remains suppressed.

These findings demonstrate that apparent robustness under aggregate accuracy or consistency metrics may conceal substantial degradation in uncertainty handling and safety-relevant behavior. They also indicate that multilingual adaptation depends not only on the inclusion of multiple languages, but on language-invariant and quality-controlled supervision.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 describes the HerHealthEval methodology, Section 4 presents the evaluation results, and Section 5 discusses the main findings and limitations.

2 Literature Review

This section reviews prior work on women’s-health LLM evaluation, multilingual and sociocultural language understanding, and clarification-centered dialogue evaluation.

2.1 Women’s-Health LLM Evaluation

Recent work has increasingly explored the use of LLMs for women’s-health communication, primarily evaluating the quality of generated responses in terms of factuality, readability, empathy, and safety [13, 3].

Domain-adapted systems such as MenstLLaMA [1] and Mai [12] demonstrate the potential of adaptation for menstrual-health dialogue. PCOS and fertility studies similarly focus on clinician-authored questions or expert judgments of generated responses [5, 2].

WHBench broadens this line of work through expert-designed safety rubrics and structured response evaluation [11]. However, these approaches generally focus on the quality of the generated answer rather than separately testing whether the model first identified the underlying concern, calibrated its urgency, and recognized when clarification was required.

2.2 Multilingual and Sociocultural Language Understanding

Multilingual healthcare evaluation has shown that semantically equivalent questions can elicit substantially different recommendations across languages. XLingEval evaluates correctness, consistency, and verifiability across English, Spanish, Chinese, and Hindi, revealing considerable cross-language variation in medical responses [7].

Subsequent studies similarly report inconsistent advice for equivalent clinical questions across languages [14].

Cross-cultural NLP further distinguishes linguistic form from cultural knowledge and communicative goals [6]. Participatory reproductive-health research likewise shows that suitable responses may require local context beyond literal translation [4]. HerHealthEval therefore limits its claims to controlled language and register variation rather than treating language as a proxy for culture.

2.3 Ambiguity, Clarification, and Understanding-Centered Evaluation

Clarification research treats recognizing uncertainty, identifying missing information, requesting additional evidence, and responding after clarification as distinct capabilities that remain challenging for current dialogue systems [15, 16].

2.4 Gap and Evaluation Requirements

The central gap is not another response benchmark, but an evaluation framework that tests whether models preserve the same underlying concern when equivalent meanings are expressed across registers and languages.

Table 1 summarizes the closest women’s-health LLM systems and illustrates this gap. Existing systems primarily evaluate educational response generation, expert-scored answer quality, or dialogue helpfulness, whereas HerHealthEval focuses on multilingual symptom-language understanding through matched semantic variants and explicit triage and clarification targets.

Such an understanding-centered evaluation framework requires four properties:

1. preserving semantic intent while varying linguistic realization across registers and languages;
2. evaluating concern interpretation, risk calibration, and recognition of constructed under-specification separately;
3. exposing under-triage and degenerate consistency rather than relying solely on aggregate accuracy; and
4. preventing contamination between adaptation and evaluation data.

HerHealthEval operationalizes these requirements through matched multilingual register variants, explicit interpretation targets, and a leakage-control protocol based on exact-match, source-provenance, and paraphrase-family exclusion.

Study	Goal	Languages	Matched forms	Triage / clarification targets
MenstLLaMA [1]	Educational answer generation	English	No	No
Mai [12]	Menstrual-health dialogue generation	English, Roman Urdu	No	No
WHBench [11]	Expert-scored clinical responses	English	No	Safety rubric; no clarification target
HerHealthEval	Symptom-language interpretation	English, French, Arabic	Yes	Yes

Table 1 Positioning against the closest women’s-health LLM systems. All three HerHealthEval languages are used for both adaptation and matched scored evaluation ($n = 540$ per language).

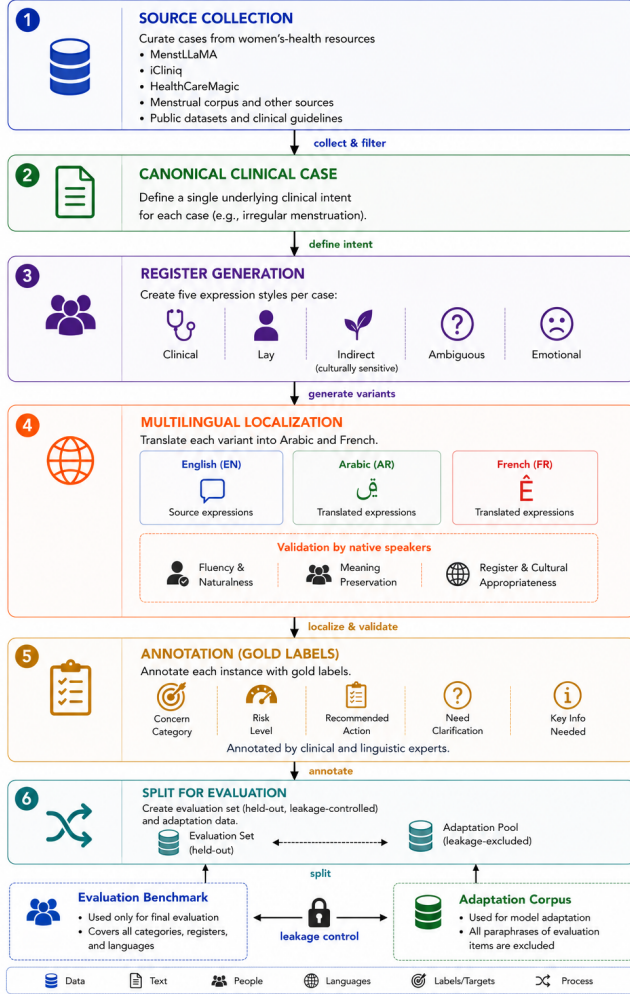


Fig. 2 HerHealthEval construction pipeline.

3 Methods

3.1 Overview of the Proposed Framework

HerHealthEval holds case identity fixed while varying linguistic realization across languages and six forms: one canonical concern plus clinical, lay, indirect or hedged, deliberately under-specified, and emotionally concerned variants. The canonical, clinical, lay, indirect, and emotional forms preserve the available clinical meaning. The under-specified form is an explicit stress test that deletes rele-

vant detail and is therefore analyzed separately rather than treated as a meaning-preserving paraphrase.

For example, *menstrual irregularity* may appear in several meaning-preserving forms, such as “I have irregular menstrual bleeding,” “my period is all over the place,” “something about my cycle feels wrong,” or “I’m worried because my cycle hasn’t been normal for months.” These forms differ in explicitness, directness, and emotional framing without changing the available clinical meaning. A separate deliberately under-specified form omits relevant information by design and is evaluated as a clarification stress test.

Predicted differences can therefore be attributed to linguistic realization (Figures 1 and 2). Each instance has structured targets for concern and risk, together with a target for recognizing constructed under-specification. The latter is not a clinician-adjudicated judgment of whether a real patient requires clarification.

3.2 Source

PCOS or hormonal and fertility questions were drawn from the HealthCareMagic-100k and iCliniq-10k patient–doctor datasets [9, 8], accessed through a Hugging Face repackaging [10].

Menstrual-health questions came from the MENST dataset [1], covering menstruation, irregular periods, premenstrual syndrome, and menopause. Its source-level paraphrase-family membership was retained for leakage control (Section 3.8).

Candidates were keyword-filtered into the three categories and cleaned of a recurring ChatDoctor text-substitution artifact. Near-duplicate, myth-only, off-topic, and non-first-person cases were removed; the remainder were balanced across categories.

Where reliable mappings existed, concern categories were grounded in NHS and NICHD guidance, supplemented by CDC fertility guidance. Cases lacking evidence for deterministic assignment were flagged rather than given inferred labels.

3.3 Canonical Clinical Case

Each case is mapped to one canonical clinical concern that anchors all variants. Thus, “I have irregular menstrual bleeding,” “my period is all over the place,” and “something about my cycle feels wrong” map to *menstrual irreg-*

ularity. Canonicalization makes linguistic variation a controlled variable while preserving intent.

3.4 Register Construction

Canonical cases are rewritten under a strict meaning-preservation constraint: wording, directness, hedging, and emotional framing may vary, but clinical evidence and concern category may not.

Template-based pilots caused lexical convergence, especially in the clinical register. The final protocol therefore applies distinctness and semantic-preservation checks, followed by human review for semantic consistency, register fidelity, and naturalness.

3.5 Multilingual Localization

Matched English cases were localized into French and Modern Standard Arabic (MSA), selected over regional dialects to support terminological consistency and controlled cross-lingual comparison. Initial translations were generated using a standardized LLM-assisted localization protocol, with explicit instructions to preserve the clinical meaning and communication register of each source instance. Automated checks were applied for structural integrity and source–target alignment, followed by native-speaker review for semantic equivalence, register fidelity, and naturalness. Arabic translations underwent an additional review by a professional Arabic–English medical translator, with particular attention to medical terminology, grammatical accuracy, and gender agreement. Final localized instances were retained only after human linguistic validation.

3.6 Annotation

Because public-health guidance mapped most cases to one broad risk category, a transparent deterministic keyword heuristic derives silver risk targets from source answer text.

For the held-out evaluation benchmark, risk is computed once per English source case and propagated by shared case identity. Matched English, French, and Arabic evaluation items therefore carry identical, language-invariant labels.

Clarification targets are assigned mechanically from the experimental construction: the deliberately underspecified form is positive and all other forms are negative. These labels test recognition of the injected information gap; they do not represent a general clinical standard for when clarification is required. They are shared across matched languages.

Both targets are silver rather than clinician-adjudicated. A post-hoc audit found that the original multilingual adaptation corpus had instead recomputed French and Arabic risk labels by applying the English-keyed heuristic directly to translated answer text. Missing English consultation terms produced language-asymmetric labels heavily skewed toward **routine**.

We retain the resulting model, M3-ML, as a documented failure case. To isolate this defect, M3-ML-RC derives risk once from the English source answer and attaches

it to matched English, French, and Arabic adaptation rows through `row_id`, producing language-invariant targets.

All other adaptation procedures are retained, isolating language-asymmetric supervision from multilingual adaptation generally.

3.7 Split for Evaluation

Localized, annotated cases are partitioned into disjoint corpora: an *adaptation corpus* for English specialization and multilingual adaptation, and an *evaluation benchmark* reserved for controlled register and language evaluation.

3.8 Leakage Control

The benchmark holds out concern families across languages and registers; the adaptation corpus provides domain exposure without revealing their instances or multilingual realizations.

Because cases may share a concern while differing only in wording, register, or language, row-level splitting can leak paraphrases, translations, or alternative realizations.

We therefore apply three exclusion rules: (i) **Exact-match exclusion**, preventing identical formulations from appearing across the two corpora; (ii) **Source-provenance exclusion**, preventing cases derived from the same source instance from crossing the split; and (iii) **Paraphrase-family exclusion**, preventing multilingual or cross-register realizations of the same canonical concern from appearing in both corpora.

Applied across all languages, these constraints make evaluation measure generalization to unseen realizations rather than memorization.

3.9 Metrics

Predefined-label metrics include concern and risk accuracy, under- and over-triage, and clarification recall and specificity.

Given class imbalance, risk accuracy is reported with its majority baseline. Under-triage is the proportion of reference **see-doctor** cases assigned **routine**; over-triage is the opposite error. Reporting both prevents aggregate accuracy from hiding safety-relevant class behavior.

Risk and category accuracy, triage, and clarification metrics use parse-valid outputs; failures are reported separately through `parse_ok`. The under-triage denominator is therefore the parse-valid subset whose reference risk is **see-doctor**.

We also report parse compliance and cross-register consistency: the fraction of case groups whose variants receive one risk or concern label. Pairwise and three-way agreement uses matched English, French, and Arabic cases. Each language contains 540 matched evaluation items with shared case identities and labels.

The model-reported `unsafe_response` flag is auxiliary. McNemar’s test compares matched per-item predictions on shared items; the M3-ML versus M3-ML-RC comparison is additionally tested on reference **see-doctor** cases using avoidance of **routine** as the outcome.

Metric	Model 1	Model 2	Model 3	Model 4	Majority
parse_ok	1.000	0.998	0.996	0.998	—
risk accuracy	0.667	0.623	0.665	0.655	0.644
under-triage ↓	0.718	0.647	0.605	0.711	—
clarif. recall	0.856	0.044	0.000	0.000	—
category accuracy	0.539	0.549	0.550	0.510	—
consistency (risk)	0.644	0.433	0.444	0.500	—
consistency (cat.)	0.156	0.344	0.378	0.411	—

Table 2 English results ($n = 540$). Model 1 is zero-shot; Model 2 uses structured English QLoRA; Model 3 adds $4\times$ clarification oversampling; and Model 4 uses joint English–French–Arabic QLoRA. “Majority” is the majority-class baseline. Lower under-triage is better; consistency requires the same label across all six forms.

4 Results

We evaluate five model configurations. Model 1 is the zero-shot Qwen3.5-9B-Instruct baseline. Model 2 uses structured English QLoRA adaptation, while Model 3 extends Model 2 with $4\times$ clarification oversampling. Model 4 uses joint English–French–Arabic QLoRA adaptation, and Model 5 applies risk-corrected multilingual re-adaptation. All five models are evaluated on the English benchmark ($n = 540$). Models 1, 4, and 5 are also evaluated on matched French and Modern Standard Arabic sets ($n = 540$ each) that preserve case identity and language-invariant silver reference labels. We additionally report the majority-class accuracy to contextualize performance under class imbalance.

4.1 Zero-shot Baseline (Model 1)

Table 2 reports results for Model 1, the zero-shot baseline, a general-purpose Qwen3.5-9B-Instruct model prompted with the structured triage schema. Model 1 achieves perfect parse compliance and a risk accuracy of 0.667, only modestly exceeding the majority-class baseline of 0.644.

The dominant safety failure is clinical under-triage. Among the 174 cases whose reference risk label is **see-doctor**, Model 1 incorrectly routes 71.8% to the lower-risk **routine** category, yielding a recall of approximately 0.282 for **see-doctor**, compared with 0.897 for **routine**. This asymmetry suggests that aggregate risk accuracy substantially understates safety-critical failure modes.

In contrast, Model 1 exhibits strong uncertainty handling, achieving a clarification recall of 0.856 and a specificity of 0.862 on the 90 cases requiring clarification. Concern-category accuracy reaches 0.539 overall, with fertility concerns proving the most difficult category (recall 0.467).

Interpretation is also sensitive to linguistic realization: risk and concern labels are consistent across all six forms in only 64.4% and 15.6% of case groups, respectively. Thus, communication style substantially influences model interpretation (RQ1; Section 1).

4.2 Consistency Masks Safety Regressions

QLoRA adaptation uses leakage-controlled corpora separated from the evaluation benchmark by the protocol in Section 3.8, with the evaluation output schema.

All adapted models retain near-perfect parse compliance (≥ 0.996) and raise English category consistency from 0.156 (Model 1) to 0.344, 0.378, and 0.411 for Models 2, 3, and 4, respectively. Model 3 provides the best adapted-model triage result, with 0.665 risk accuracy and the lowest under-triage rate among the adapted English models (0.605), compared with a risk accuracy of 0.667 and an under-triage rate of 0.718 for Model 1.

Uncertainty handling deteriorates: clarification recall falls from 0.856 (Model 1) to 0.044 (Model 2) and 0.000 (Models 3 and 4). Thus, oversampling improves English triage but not clarification-seeking. A recall of 0.000 means that Models 3 and 4 did not identify any of the 90 deliberately under-specified cases as requiring clarification. The failure of clarification oversampling to recover clarification recall suggests that increased exposure alone is insufficient to preserve uncertainty-sensitive behavior during adaptation and that clarification may require an explicit training objective.

McNemar’s tests find Model 2 worse than Model 1 on risk correctness ($p = 0.033$), but not category ($p = 0.707$) or clarification correctness ($p = 0.302$). Models 3 and 4 do not differ significantly from Model 1 on matched per-item accuracy, locating the main adaptation effects in safety-sensitive and group-level metrics.

Model 4 has the highest adapted-model category consistency (0.411) but does not preserve Model 3’s triage gain. We next examine its cross-lingual behavior and risk-corrected re-adaptation.

4.3 Cross-Lingual Safety Regressions

Table 3 compares Models 1, 4, and 5 on the matched sets. Modest aggregate differences mask severe cross-lingual safety regressions and their recovery.

Model 4 leaves English under-triage nearly unchanged (0.718→0.711) but raises it to 0.994 in both French and Arabic (from 0.632 and 0.638). This corresponds to approximately 173 of 174 reference **see-doctor** cases being incorrectly assigned the lower-risk **routine** label in both non-English languages. For Model 4, clarification recall falls to 0.000 in English and French and to 0.011 in Arabic. This failure is multilingual, whereas the near-total risk collapse is concentrated in French and Arabic.

The resulting consistency gains are misleading. French and Arabic risk consistency rises from 0.544 to 0.978 and 0.456 to 0.967 because Model 4 nearly always predicts **routine**. French–Arabic agreement reaches 0.991 through shared collapse, while three-way agreement falls from 0.846 to 0.804. Degenerate consistency is not safe understanding.

Model 5 reduces under-triage below Model 1 to 0.572 (French), 0.558 (Arabic), and 0.590 (English). On reference **see-doctor** cases, it corrects 74 French misses with one reversal ($b = 1$, $c = 74$; McNemar $p = 4.0 \times 10^{-21}$), and 75

Metric	English			French			Arabic		
	Model 1	Model 4	Model 5	Model 1	Model 4	Model 5	Model 1	Model 4	Model 5
parse_ok	1.000	0.998	0.994	1.000	0.998	0.991	1.000	0.998	0.989
risk accuracy	0.667	0.655	0.663	0.685	0.646	0.677	0.674	0.646	0.684
under-triage ↓	0.718	0.711	0.590	0.632	0.994	0.572	0.638	0.994	0.558
clarif. recall	0.856	0.000	0.000	0.811	0.000	0.000	0.889	0.011	0.000
category accuracy	0.539	0.510	0.531	0.561	0.512	0.523	0.533	0.531	0.545
consistency (risk)	0.644	0.500	0.367	0.544	0.978	0.467	0.456	0.967	0.422

Table 3 Results for Model 1 (zero-shot), Model 4 (multilingual QLoRA), and Model 5 (risk-corrected multilingual QLoRA) on 540 matched items per language. Model 4 shows near-total non-English under-triage and artificially high consistency through collapse onto **routine**. Corrected risk supervision reverses the triage collapse but does not recover clarification-seeking.

Arabic misses with no reversal ($b = 0$, $c = 75$; $p = 5.3 \times 10^{-23}$). Here, b denotes cases corrected in the comparison direction and c denotes cases incorrectly changed in the opposite direction.

Aggregate risk accuracy understates this gain because Model 5 trades some **routine** accuracy for correct **see-doctor** escalation. Clarification recall remains 0.000 in every language, so risk correction does not repair clarification failure.

5 Discussion

The results answer the three research questions provisionally. For RQ1, equivalent concerns across registers yield inconsistent predictions, showing sensitivity to linguistic realization. For RQ2, multilingual evaluation of Model 4 reveals near-total French and Arabic under-triage under the flawed language-asymmetric supervision condition, hidden by aggregate accuracy and high, collapse-driven consistency. For RQ3, adaptation improves selected consistency and triage measures but suppresses clarification-seeking; source-derived, language-invariant risk labels remove the additional non-English risk collapse.

Importantly, clarification recall measures recognition of deliberately constructed under-specification rather than a clinician-adjudicated need for clarification. The near-zero recall should therefore be interpreted as a failure to recognize the injected information gap, rather than as direct evidence of clinical unsafety.

Model 5 therefore implicates the adaptation-label pipeline rather than multilingual adaptation alone. However, it does not establish clinical safety. Robust multilingual healthcare evaluation must separate understanding, uncertainty handling, and risk calibration; adaptation requires language-invariant, quality-controlled supervision.

Limitations

HerHealthEval is an evaluation framework rather than a fixed-scale benchmark. This study covers three women’s-health categories in English, French, and Modern Standard Arabic using one multilingual model family and a limited set of adaptation configurations.

The reported uncertainty tests treat matched items as individual pairs. They do not provide cluster-aware confidence intervals over source cases. Consequently, inferential claims should be read as exploratory.

Future work will focus on validating the framework across diverse clinical and linguistic settings, while incorporating clinician-adjudicated annotations and uncertainty-aware dialogue mechanisms to improve the robustness and reliability of the system.

Ethical Considerations

This study evaluates multilingual women’s-health language understanding under controlled conditions and does not provide medical advice or support clinical decision-making.

Public patient–doctor dialogue datasets are combined with machine-generated multilingual variants reviewed for semantic preservation and register fidelity. Analyses concern aggregate behavior rather than individual patients or clinical outcomes.

Any future release will follow appropriate privacy, governance, and de-identification procedures. To preserve double-blind review, the benchmark and adaptation data are withheld from this submission. Following acceptance and institutional approval, the framework, annotation protocol, and supporting resources are intended for release.

Acknowledgments

The authors thank the SD-AI ERA, School of Computing, Queen’s University, for academic supervision and research support. Large language models were used as experimental subjects in the evaluation reported in this paper; the manuscript’s claims and final text were reviewed and approved by the authors.

Competing Interests

The authors have no conflicts of interest to declare that are relevant to the content of this paper.

Ethics Approval

This study used publicly available, de-identified patient–doctor dialogue data and did not recruit human participants or collect new personal data. Accordingly, institutional ethics approval and informed consent were not applicable to the analyses reported here.

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